You are supposed to read your textbook but I will cover some of it for you

Chapters 6 & 7

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Course

ECO 375

Ch6

Multiple Regression · Further Issues

Ch 7

Multiple Regression · Qualitative Information

Materials Introductory Economitrics A Modern Approach by Jeffrey M. Wooldridge

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Units & Scaling — What Really Changes?

Changing the unit of y

If $y^* = a \cdot y$, then for any regressor j:

- $\hat{\beta}_j^* = a \cdot \hat{\beta}_j$
- $se(\hat{\beta}_j^*) = a \cdot se(\hat{\beta}_j)$
- **t-stats unchanged**

Changing the unit of x_k

If $x_k^* = b \cdot x_k$:

- $\hat{\beta}_k^* = \hat{\beta}_k / b$
- $se(\hat{\beta}_k^*) = se(\hat{\beta}_k)/b$
- Other coefficients & SEs *unchanged*

Rulers change units, not relationships.

Q2. Log-Level — Exact % Effect (Rooms → Price)

Model

$ln(price) = \alpha + \beta \cdot rooms + u$ with $\hat{\beta} = 0.306$

Percentage Effect Calculations

(a) **Approximate** % effect of +1 room: $100\beta \approx 30.6\%$

(b) **Exact** % effect: $100 \times (\exp(0.306) - 1) \approx 35.8\%$

(c) For **−1 room**: $100 \times (1 - \exp(-0.306)) \approx 26.4\%$ decrease

Interpretation

Adding one additional room is associated with approximately a 35.8% increase in home price, holding other factors constant.

When $|\beta| > 0.1$, the gap between approximate and exact formulas grows. Always use the exact formula for dummy variables and larger effects.

Functional Form — Logarithms (4 Canonical Forms)

Forms

Model	Form	Interpretation (small changes)
Level–Level	`y = α + βx`	+1 in x ⇒ β units in y
Log–Level	`ln y = α + βx`	+1 in x ⇒ ≈ **100β%** in y
Level–Log	`y = α + β ln x`	+1% in x ⇒ β/100 units in y
Log–Log	`ln y = α + β ln x`	+1% in x ⇒ **β%** in y (elasticity)

Usage Notes

- Use logs for **positive, skewed variables** (wages, sales, populations); can reduce heteroskedasticity.
- Avoid logging years (education, age) or inherently linear percentages when interpretation suffers.
- For dummies in log-DV models: use exact percentage $100 \times (e^\beta - 1)\%$ rather than approximation.
- Double-log models give **elasticities** — useful for policy analysis and comparing across different scales.

Approx vs Exact % Effect (Log-DV)

Formulas

- **Approximate:** $100\beta\%$
- **Exact:** $100 \times (\exp(\beta) - 1)\%$

For larger values of $|\beta|$, the approximation becomes increasingly inaccurate:

- $\beta = 0.1$: approx 10%, exact 10.5%
- $\beta = 0.3$: approx 30%, exact 35.0%
- $\beta = 0.5$: approx 50%, exact 64.9%

Guidance

- The gap between approximate and exact values grows as $|\beta|$ increases (noticeable for $|\beta| \gtrsim 0.1$).
- **Always use the exact formula** for dummy jumps ($0 \rightarrow 1$) or large effects.
- For decreases (e.g., $1 \rightarrow 0$): $100 \times (1 - \exp(\beta))\%$
- Example: If $\beta = -0.3$, approximate is -30%, but exact decrease is -25.9%

For log-dependent variable models, exact percentage interpretation matters.

Standardized Coefficients

What Are They?

Standardize variables to SD units:

- $y_s = (y - \bar{y})/s_y$
- $x_{js} = (x_j - \bar{x}_j)/s_{x_j}$

Interpretation: A 1-SD increase in x_j is associated with a $\tilde{\beta_j}$ SD change in y.

When to Use

- **Comparing magnitudes** across variables with different units
- **Communication** with non-specialists (size intuition)
- Assessing **relative importance** of variables

Best Practices

- Report *both* raw and standardized coefficients
- Avoid mechanical interpretations; consider theoretical importance
- Remember that statistical significance doesn't change

Common Pitfalls

- Not helpful for dummies (already have natural scale)
- Doesn't solve multicollinearity problems
- Misleading for variables with unusual distributions

Quadratics — Diminishing/Increasing Returns

The Model

Basic quadratic specification:

$y = \alpha + \beta_1 x + \beta_2 x^2 + u$

Where:

- β_1 captures linear effect
- β_2 captures curvature

Marginal Effects

The effect of a small change in x:

$\partial y/\partial x = \beta_1 + 2\beta_2 x$

Key insight: The effect of x *depends on the level of x.*

If $\beta_2 < 0$: diminishing returns (slowing gains)

If $\beta_2 > 0$: increasing returns (accelerating gains)

Finding the Turning Point

When $\beta_2 < 0$, the function reaches a maximum at:

$x^* = -\beta_1/(2\beta_2)$

When $\beta_2 > 0$, this gives a minimum point.

Example: If returns to experience peak at x years, this formula tells us where.*

Example: Wage–Experience

Model: $\hat{y} = \alpha + 0.286 \cdot exp - 0.0055 \cdot exp^2$

- At exp=1: return is 0.275
- At exp=10: return falls to 0.176
- Peaks at 26 years (then flattens/falls)

Captures the empirical pattern that wage gains from additional experience diminish over time.

Q3. Quadratic Turning Point (Experience & Wages)

Model

$\hat{y} = \alpha + 0.286 \cdot exp - 0.0055 \cdot exp^2$

Tasks

1. Calculate the marginal return at **exp=1** and **exp=10**
2. Find the **turning point** where returns peak
3. Interpret the economic meaning

Solution

(a) Marginal return: $\partial y / \partial exp = 0.286 - 0.011 \cdot exp$

At exp=1: $0.286 - 0.011(1) = 0.275$

At exp=10: $0.286 - 0.011(10) = 0.176$

(b) Turning point: $x^* = -\beta_1/(2\beta_2) = -0.286/(2 \times -0.0055) \approx 26.0$ years

(c) Interpretation: Wage returns to experience **diminish over time** ($\beta_2 < 0$) and peak at 26 years, after which additional experience provides decreasing returns.

Key insight: Quadratics capture nonlinear relationships where effects strengthen or weaken as variables change.

Q4. Interactions — Bedrooms × Square Footage

Model

$ln(price) = \alpha + \beta_1 \cdot bdrms + \beta_2 \cdot sqrft + \beta_3(bdrms \times sqrft) + u$

Tasks

1. Interpret the effect of **one more bedroom** at a given house size
2. How would you test whether the effect of bedrooms depends on house size?

Solution

(a) Effect of +1 bedroom at size S: $100 \times (\beta_1 + \beta_3 \cdot S)\%$

Interpretation: If $\beta_3 > 0$, bedrooms add **more value** in larger houses (complementarity)

(b) Test the null hypothesis: $H_0\colon \beta_3 = 0$ using a t-test

Rejecting implies the effect of bedrooms varies with square footage (slope heterogeneity)

Key insight: Interactions allow effects to vary with context. They transform "it depends" intuitions into testable hypotheses.

Interactions — Effects Depend on Context

Continuous × Continuous

Model: $y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 x_2) + u$

- Effect of x_1: $\beta_1 + \beta_3 x_2$ (depends on x_2)
- Positive β_3 → variables are complements
- Negative β_3 → variables are substitutes

Example: returns to experience may depend on education level

Dummy × Continuous

Model: $y = \alpha + \beta_1 x + \beta_2 D + \beta_3 (D \times x) + u$

- Effect of x when $D=0$: β_1
- Effect of x when $D=1$: $\beta_1 + \beta_3$
- Allows **different slopes** by group

Example: returns to education may differ by gender/region

Dummy × Dummy

Model: $y = \alpha + \beta_1 A + \beta_2 B + \beta_3 (A \times B) + u$

- β_3 captures **subgroup deviations** beyond additive effects
- Without interaction: effects are purely additive
- Test H_0: $\beta_3 = 0$ to assess interaction

Example: married×female effect beyond individual components

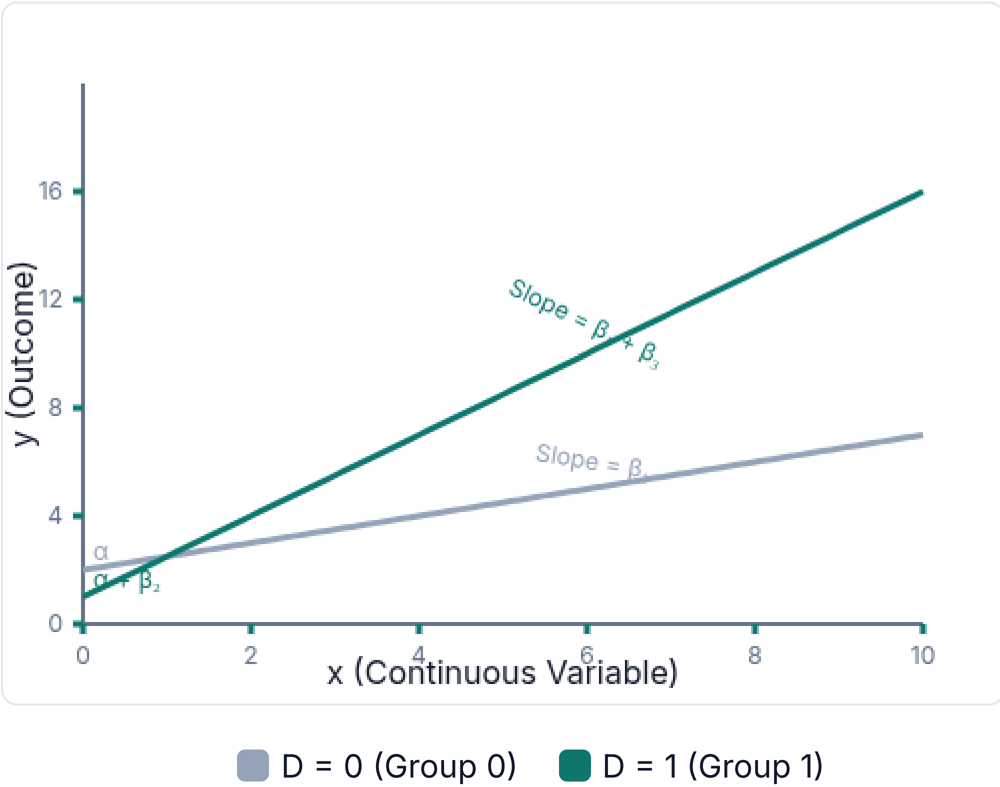
Visual Intuition

- **No interaction**: parallel shifts (pure level effect)
- **Dummy×continuous**: different slopes across groups
- **Continuous×continuous**: crossover effects (context-dependent)

Always report effects at meaningful values of the interacted variables

Interactions — Different Slopes by Group (Dummy × Continuous)

Visualization



Dummy × Continuous: Different slopes across groups

Interpretation

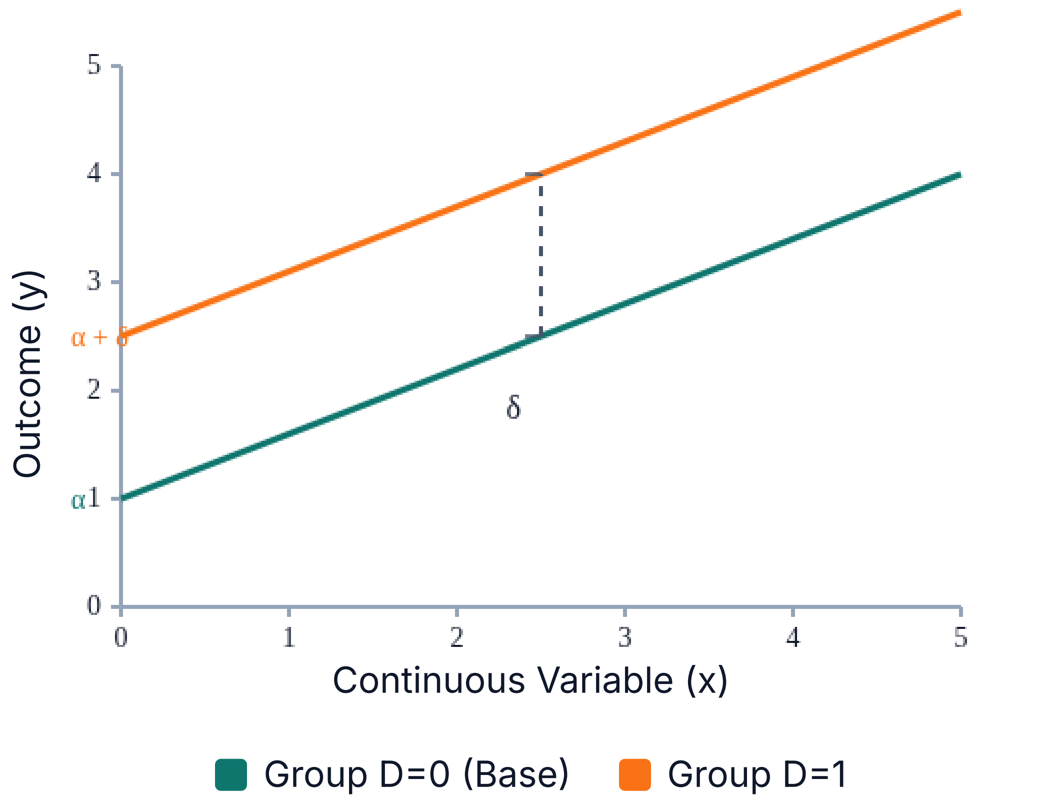
$y = \alpha + \beta_1 x + \beta_2 D + \beta_3(D \times x) + u$

- Slope when D = 0: β_1
- Slope when D = 1: $\beta_1 + \beta_3$
- Intercept shift for D = 1: β_2
- **Key insight:** $\beta_3 \neq 0$ implies the effect of x depends on group membership
- **Test for slope heterogeneity:** H_0: $\beta_3 = 0$ (t-test); include interaction if rejected

Example: Returns to education differ by gender; wage effect of experience varies by location; etc.

Interactions — Parallel Shifts (Dummy Only: Different Intercepts)

Visual Representation



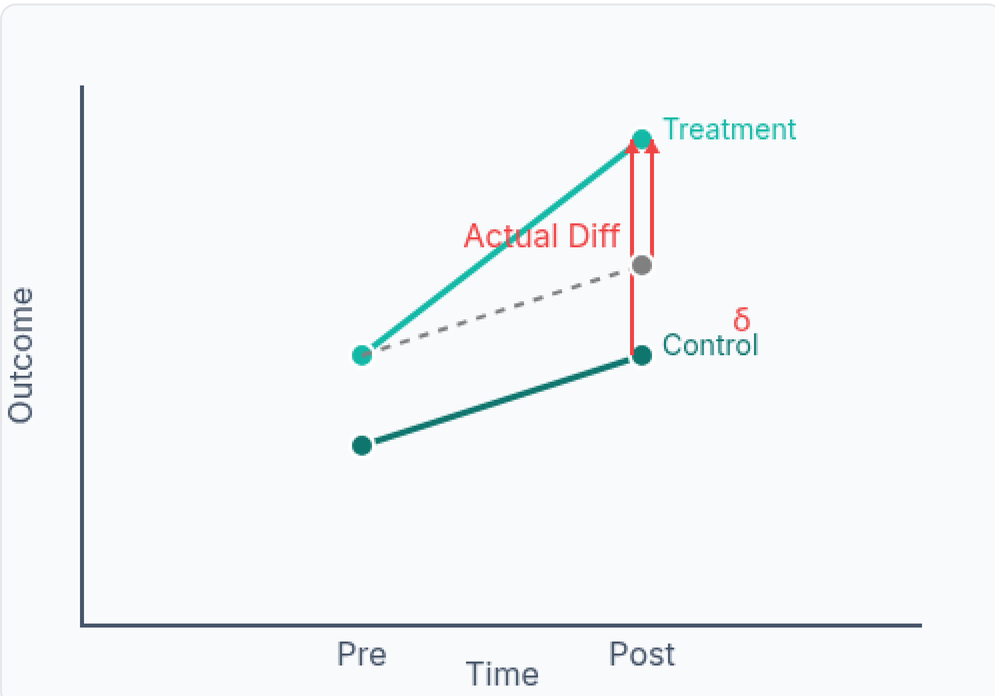
Key Properties

- Model: $y = \alpha + \beta x + \delta D + u$ (no interaction term)
- Slope (both groups): β (parallel lines)
- Intercept shift for D=1: δ (group difference holding x fixed)
- Interpretation: δ is the **constant** difference between groups at any value of x
- When appropriate: group differences are level shifts, not slope changes
- Test: If interaction H_0: $\beta_3 = 0$ is not rejected, prefer this simpler model

Parallel shifts represent additive effects of the dummy variable without slope changes.

Interactions — Subgroup Deviations (Dummy × Dummy, DiD Style)

Difference-in-Differences Visualization



The δ interaction term captures the "extra effect" beyond additive group and time effects

Interpretation

- Model (DiD): $y = \alpha + \tau\ Treat + \pi\ Post + \delta\ (Treat{\times}Post) + u$
- δ (Treat×Post) = Difference-in-Differences effect
- Interpretation: δ captures the treatment effect beyond additive group and time effects
- Key assumption: Parallel trends (in absence of treatment, groups would change similarly)
- Test: H_0: $\delta = 0$ (t-test); consider pre-trend checks if panel data available

2×2 Groups & an Interaction

Table (Effects relative to base A=0,B=0)

	Group B=0	Group B=1
A=0	Base mean	+ β_B
A=1	+ β_A	+ β_A + β_B + $\boldsymbol{\beta_{AB}}$

Interpretation

- β_{AB} captures the extra deviation beyond additive effects
- Use when subgroup differs more/less than sum of parts
- Test H_0: $\beta_{AB} = 0$ to assess interaction

The interaction term reveals when combined effects exceed or fall short of what individual effects would predict.

Adjusted R^2 & Nonnested Choices

Formula & Intuition

$$\bar{R}^2 = 1 - ((n - 1)/(n - k - 1)) \times (1 - R^2)$$

- Rewards fit, penalizes unearned complexity
- Adding noise variables may raise R^2 but often *lowers* $\bar{R}^2$
- Punishes each additional regressor (k) that doesn't contribute enough explained variation

When you're comparing models with the same dependent variable, saying "choose the model with the higher adjusted R^2" is equivalent to saying "choose the model with the smaller estimated error variance."

Less strict compared to hypothesis testing

Comparison Rules

- Only compare across nonnested models when the DV form is **identical** (e.g., log(y) vs. log(y))
- Not valid for comparing models with different dependent variables (e.g., y vs. log(y))
- Practice: Prefer theory + diagnostics; use $\bar{R}^2$ as a pragmatic tie-breaker
- A model with fewer variables and similar $\bar{R}^2$ is generally preferable (parsimony)

It's only useful in choosing between two models of the same dependent variable. If there are more than two choices, we should use something called an information criterion or cross-validation (which we won't be covering in this course

Q5. Adjusted R^2 — Model Choice

Setup

Two models for the same sample with dependent variable = ln(y):

Model 1: log-log	Model 2: log-level + extra controls
$R^2 = 0.52$	$R^2 = 0.54$
$k_1 = 5$ regressors	$k_2 = 9$ regressors

Sample size $n = 800$

Task

Compute adjusted R^2 for both models and determine which specification is preferred.

Solution

Using the adjusted R^2 formula: $\bar{R}^2 = 1 - ((n-1)/(n-k-1))\cdot(1-R^2)$

M1: $\bar{R}^2(M1) = 1 - ((800-1)/(800-5-1))\cdot(1-0.52) \approx 1 - (799/794)\cdot 0.48 \approx 1 - 0.484 \approx 0.516$

M2: $\bar{R}^2(M2) = 1 - ((800-1)/(800-9-1))\cdot(1-0.54) \approx 1 - (799/790)\cdot 0.46 \approx 1 - 0.465 \approx 0.535$

Decision: Since $\bar{R}^2(M2) > \bar{R}^2(M1)$, we would prefer Model 2 based on adjusted R^2.

Key insight: The higher R^2 of Model 2 is not just due to adding variables; the improvement survives the adjusted R^2 penalty for complexity. Still, confirm with theory and other diagnostics.

Big Picture — Chapter 6 (Multiple Regression: Further Issues)

Scaling ≠ Substance

- Change unit of y → all coefficients & SEs scale by same factor → t unchanged.
- Change unit of an x → that coefficient & SE rescale; others unaffected.

Compare Apples via SDs

Standardized coefficients: "a 1-SD ↑ in x associates with $\tilde{\beta}$ SD in y".

Functional Forms

- Logs → percent changes;
- Quadratics → diminishing/increasing returns;
- Interactions → effects depend on context.

Adjusted R^2

Penalizes bloat. Use as a pragmatic comparator when DV form is identical.

Qualitative Information via Dummies — Basics

- **Single dummy** (intercept shift): $y = \alpha + \delta D + \gamma' X + u$

  δ is the mean difference between D=1 and D=0 groups, *holding X fixed*.

- **Log-DV with dummy**: $\ln y = \alpha + \delta D + \gamma' X + u$

  Exact percentage effect: $100 \times (\exp(\delta) - 1)\%$

  Always use exact formula for dummy jumps (0→1) or large effects.

- **Multiple categories**: with *k* groups, include *k−1* dummies

  Choose a clear base group and state it explicitly in tables and text.

  Example: base = "single men"; dummies for married men, married women, single women.

Q6. Dummies — Gender Gap with Log Wages

Setup

Model: $ln(wage) = \alpha + \delta \cdot female + \gamma' X + u$ with $\hat{\delta} = -0.297$

Tasks

1. Calculate the **approximate** gender wage gap
2. Calculate the **exact** percentage gap
3. Add education slope heterogeneity by gender and interpret

Solution

(a) Approximate gap: $100\hat{\delta} \approx -29.7\%$

(b) Exact gap: $100 \times (\exp(-0.297) - 1) \approx -25.7\%$

Interpretation: Women earn approximately 25.7% less than comparable men (controls X held fixed)

(c) Include $female \times education$ interaction term

Return to education for women: $\beta_{edu} + \beta_{int}$

Return to education for men: β_{edu}

A positive β_{int} means **larger** education returns for women

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Key insight: Dummies in log models require exact percentage conversion; interactions with dummies reveal group-specific slopes.

Big Picture — Chapter 7 (Qualitative Information / Dummies)

Dummy Variables Basics

- Qualitative factors → **dummy variables** (0/1) that shift intercepts (group differences).
- When DV is in logs, a dummy's exact percentage effect: $100 \times (\exp(\delta) - 1)\%$.

Multiple Categories

- With k categories, include $k-1$ dummies.
- Choose a clear **base group** for interpretation.
- State the baseline (omitted) group explicitly in all analyses.

Dummy × Dummy Interactions

- Captures **subgroup contrasts** (e.g., married×female).
- Isolates effects beyond the sum of separate group shifts.
- Intercept = base group; coefficients = differences vs base.

Dummy × Continuous Interactions

- Allows **slope heterogeneity** across groups.
- Example: returns to education differ by gender/region.
- Differences in slopes = differences in marginal effects.

Take-away

Scale & Interpretation Issues

- Weird magnitudes? Check **units**; rescale to interpretable units.
- Big coefficients in log models? Use **exact % conversion**: $100 \times (\exp(\beta) - 1)\%$.

Nonlinearity Concerns

- Suspect curvature? Add $\mathbf{x^2}$; test β_2.
- Context matters? Add **interactions** (dummy×X or X×X).

Model Selection

- Too many controls? Compare $\bar{R}^2$ but only with same DV form.
- Prefer theory and diagnostics; let $\bar{R}^2$ break ties, not dictate choices.

Group Differences

- Groups differ? Use **dummies**; always state the base group.
- With log-y, remember to convert to **exact %** for proper interpretation.